Notes: Baker & Wurgler (Journal of Finance, 2006)

Investor Sentiment and the Cross-Section of Stock Returns

Contents

1 Big picture

- **Question.** Does investor sentiment affect the cross-section of stock returns, not just the level of the aggregate market?
- **Main idea.** Sentiment matters most for stocks whose valuations are subjective and whose mispricing is hard to arbitrage: small, young, volatile, unprofitable, non-dividend-paying, extreme-growth, and distressed firms.
- **Mechanism.** A sentiment shock creates demand pressure. If arbitrage were frictionless, rational traders would offset it. When valuation is subjective, idiosyncratic risk is high, and shorting/trading is costly, the shock can move prices and predict future returns.
- **Empirical strategy.** Build an annual sentiment index from six proxies, then ask whether monthly returns on characteristic-sorted portfolios depend on beginning-of-period sentiment.
- **Core finding.** When sentiment is high, speculative/hard-to-arbitrage stocks earn low subsequent returns relative to safer stocks. When sentiment is low, these same stocks earn high subsequent returns or the pattern reverses.
- **Why it matters.** Several characteristics that look weak or ambiguous unconditionally become powerful after conditioning on sentiment.
- **Risk alternative.** Conditional beta tests and earnings-announcement tests make a pure risk explanation less convincing.
- **Takeaway.** Sentiment is not only a market-wide variable. It reshapes the cross-section by changing the relative pricing of speculative versus bond-like stocks.

2 Data and measurement

2.1 Sentiment proxies

The sentiment index is constructed from six annual proxies. Each proxy is imperfect alone, so the authors extract a common component using principal components analysis.

- **CEFD:** value-weighted closed-end fund discount. High discounts indicate low investor sentiment, so CEFD enters the index with a negative sign.
- **TURN:** detrended log NYSE turnover. In a market with short-sale constraints, high liquidity/turnover may reflect optimistic participation.
- **NIPO:** number of IPOs. Hot IPO markets are treated as high-sentiment episodes.
- **RIPO:** average first-day IPO return. High IPO initial returns proxy for investor enthusiasm.
- **S:** equity share in new issues, measured as gross equity issuance divided by gross equity plus long-term debt issuance.
- P^{D-ND}**:** dividend premium, the log valuation difference between dividend payers and nonpayers.

Interpretation: The composite index is designed to isolate the common time-series component behind financing conditions, turnover, IPO markets, closed-end fund discounts, and dividend demand. The orthogonalized index removes macroeconomic conditions to reduce concern that sentiment is just a business-cycle proxy.

2.2 Firm characteristics

The paper uses CRSP-Compustat common stocks from 1962 to 2001 and matches fiscal-year accounting data to July-to-June monthly returns in the Fama-French tradition.

- **Size and age:** market equity and years since first CRSP appearance.
- **Risk:** total return volatility from the prior year.
- **Profitability:** positive earnings and earnings-to-book equity.
- **Dividend policy:** whether the firm pays dividends and the dividend-to-book ratio.
- **Tangibility:** PPE/assets and R&D/assets.
- **Growth/distress:** book-to-market, external finance over assets, and sales growth deciles.

Interpretation: These characteristics proxy for speculative appeal and limits to arbitrage. The same characteristics that make a firm exciting to optimists often make the stock hard to value and expensive to arbitrage.

2.3 Return portfolios

- Each month, the authors form equal-weighted characteristic decile portfolios using NYSE breakpoints.
- Equal weighting is intentional: the theory predicts sentiment effects should be strongest among small stocks, and value weighting would hide much of the action.
- The main comparisons are high-versus-low decile spreads, medium-versus-extreme spreads, and high-minus-medium or medium-minus-low spreads for U-shaped variables such as book-to-market, external finance, and sales growth.

Interpretation: The paper does not ask whether sentiment predicts the market. It asks whether sentiment predicts which types of firms outperform or underperform.

3 Models and empirical specifications

3.1 Two channels for sentiment

The paper describes two equivalent empirical channels: cross-sectional variation in sentiment demand and cross-sectional variation in arbitrage limits.

$$\text{Mispricing} = \text{sentiment-driven demand shock} \times \text{limits to arbitrage}.$$

Interpretation: The variables most exposed to sentiment demand are often the same variables that make arbitrage hard: high idiosyncratic risk, young age, low profitability, extreme growth, distress, and short-sale constraints.

3.2 Composite sentiment index

$$\begin{aligned} \text{SENTIMENT}_t = & -0.241 CEFD_t + 0.242 TURN_{t-1} + 0.253 NIPO_t && (1) \\ & + 0.257 RIPO_{t-1} + 0.112 S_t - 0.283 P_{t-1}^{D-ND}. && (2) \end{aligned}$$

$$
\begin{aligned}
\text{SENTIMENT}_t^{\perp} = & -0.198 CEFD_t^{\perp} + 0.225 TURN_{t-1}^{\perp} + 0.234 NIPO_t^{\perp} \qquad (3)\\
& + 0.263 RIPO_{t-1}^{\perp} + 0.211 S_t^{\perp} - 0.243 P_{t-1}^{D-ND,\perp}. \qquad (4)
\end{aligned}
$$

Interpretation: The signs line up with investor sentiment: high IPO activity, high IPO first-day returns, high turnover, and high equity issuance indicate high sentiment; high closed-end fund discounts and high relative valuations of dividend payers indicate lower demand for speculative stocks.

3.3 Conditional portfolio sorts

$$
E[R_{j,t+1} \mid \text{decile } j, \text{SENTIMENT}_t^{\perp} > 0] - E[R_{j,t+1} \mid \text{decile } j, \text{SENTIMENT}_t^{\perp} < 0].
$$

Interpretation: A high-sentiment period should have low subsequent returns for speculative/hard-to-arbitrage deciles because those stocks are relatively overpriced at the start of the period.

3.4 Time-series regressions

$$
R_{High,t}^{X} - R_{Low,t}^{X} = c + d\text{SENTIMENT}_{t-1} + \beta RMRF_t + sSMB_t + hHML_t + mUMD_t + u_t.
$$

Interpretation: The coefficient d is the central estimate. For example, if High–Low is high-volatility minus low-volatility, a negative d means high-volatility stocks underperform low-volatility stocks after high sentiment.

3.5 Conditional beta test

$$
R_{High,t}^{X} - R_{Low,t}^{X} = c + d\text{SENTIMENT}_{t-1} + \beta(e + f\text{SENTIMENT}_{t-1}) RMRF_t + u_t.
$$

Interpretation: If return predictability is just changing market beta, the interaction coefficient f should explain the same patterns. The paper finds little support for this explanation.

3.6 Earnings-announcement test

$$
CAR_{Decile,t}^{X} = c + d\text{SENTIMENT}_{t-1}^{\perp} + u_t.
$$

Interpretation: If sentiment reflects overly optimistic or pessimistic earnings expectations, announcement returns should partially correct those errors. This test provides a lower bound because not all valuation errors are corrected at earnings announcements.

4 Identification and empirical design

4.1 What identifies the effect

- The identifying variation is annual sentiment at the start of the period interacted with cross-sectional firm characteristics.
- The comparison is not high-sentiment versus low-sentiment aggregate market returns. It is which stocks outperform conditional on sentiment.

- The paper looks for sign reversals across sentiment states, which are harder to reconcile with fixed risk premia.
- Controls include the market factor, SMB, HML, and momentum in the regression tests.

4.2 Why the design is credible

- The sentiment index is built before the return period, which avoids contemporaneous mechanical return measurement.
- The orthogonalized sentiment index removes macroeconomic conditions such as industrial production, consumption growth, employment growth, and recessions.
- The same qualitative patterns appear in sorts, regressions, extended earlier samples, and earnings-announcement evidence.
- The strongest effects appear exactly where the theory says sentiment should matter: subjective valuation and limited-arbitrage stocks.

4.3 Remaining limits

- The sentiment index is a proxy, not a direct observation of beliefs.
- The joint-hypothesis problem remains: return predictability can always be recast as a complicated risk model unless rejected by further restrictions.
- The sample covers a specific historical period, 1960s through the Internet bubble, so out-of-sample validation is important.
- Equal-weighted results are economically important for the theory but may not map one-for-one into large-cap investable portfolios.

5 Tables and figures

5.1 Figure 1 and Figure 2: sentiment over time and conditional return shapes

Read: Figure 1.

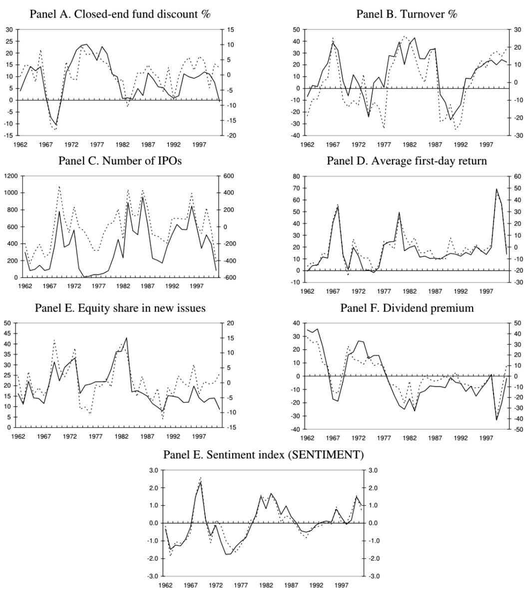
- The six sentiment proxies move together around recognizable speculative episodes: late-1960s speculation, the mid-1970s low-sentiment period, the rising sentiment of the late 1970s and 1980s, and the Internet bubble.
- The raw and macro-orthogonalized versions are visually similar, suggesting that the index is not merely a business-cycle variable.
- The final panel combines the proxies into the standardized SENTIMENT index.

Read: Figure 2.

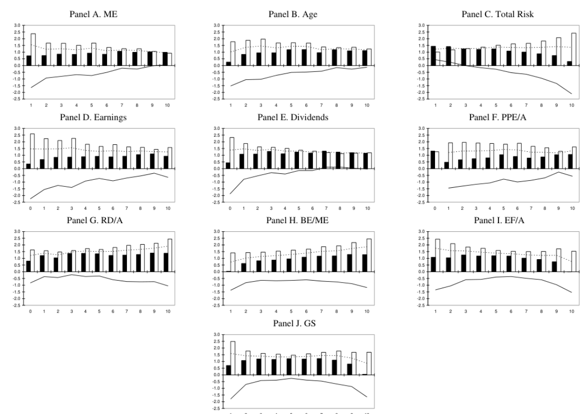
- The plots translate Table III into visual conditional-return curves across characteristic deciles.
- Small, young, volatile, unprofitable, non-dividend-paying, and extreme growth/distress portfolios are most sensitive to sentiment.
- The curves show that conditional patterns are often nonmonotonic or U-shaped: both extreme growth and distress can be sentiment-sensitive.

Economic message: Figure 1 validates the measurement strategy: several independent proxies identify similar sentiment episodes. Figure 2 shows the economic content of the paper: sentiment changes the slope and shape of the cross-section, not just the market level.

Additional interpretation: The key point in Figure 2 is not that every characteristic has a simple monotonic premium. Instead, sentiment predicts which regions of the characteristic space are overvalued or undervalued. For variables such as book-to-market, external finance, and sales growth, both extremes can be hard to value, so a single high-minus-low spread can miss the central result.



(a) Investor sentiment proxies and composite index, 1962–2001.



(b) Conditional future returns across characteristic deciles.

5.2 Tables I and II: return/characteristic data and sentiment data

Read: Table I.

- Table I defines the return variables and the firm characteristics used for portfolio sorts.
- The characteristics are organized around size/age/risk, profitability, dividends, tangibility, growth opportunities, and distress.
- The table clarifies that the paper studies common stocks from CRSP-Compustat and matches accounting data to later monthly returns.

Read: Table II.

- Table II reports summary statistics and correlations for the six sentiment proxies and the two composite sentiment indexes.
- The proxies are not identical, but they share enough common variation to justify a principal-component index.
- The orthogonalized measures retain the expected signs and relationships after macro controls.

Economic message: Tables I and II establish the measurement foundation. The return tests are only meaningful because the characteristics are well-defined and because sentiment is measured from multiple noisy but related proxies.

Additional interpretation: Table II is especially important because any single sentiment proxy could be criticized. The principal-component approach reduces reliance on one narrative such as IPO activity or closed-end fund discounts. The fact that orthogonalization does not overturn the index makes the sentiment interpretation more plausible.

Table I

Summary Statistics, 1963–2001

Panel A summarizes the returns variables. Returns are measured monthly. Momentum (*MOM*) is defined as the cumulative return for the 11-month period between 12 and 2 months prior to t. Panel B summarizes the size, age, and risk characteristics. Size is the log of market equity. Market equity (*ME*) is price times shares outstanding from CRSP in the June prior to t. Age is the number of years between the firm's first appearance on CRSP and t. Total risk (σ) is the annual standard deviation in monthly returns from CRSP for the 12 months ending in the June prior to t. Panel C summarizes profitability variables. The earnings-book equity ratio is defined for firms with positive earnings. Earnings (E) is defined as income before extraordinary items (Item 18) plus income statement deferred taxes (Item 50) minus preferred dividends (Item 19). Book equity (*BE*) is defined as shareholders equity (Item 60) plus balance sheet deferred taxes (Item 35). We also report an indicator variable equal to one for firms with positive earnings. Panel D reports dividend variables. Dividends (D) are equal to dividends per share at the ex date (Item 26) times shares outstanding (Item 25). We scale dividends by assets and report an indicator variable equal to one for firms with positive dividends. Panel E shows tangibility measures. Plant, property, and equipment (Item 7) and research and development (Item 46) are scaled by assets. We only record research and development when it is widely available after 1971; for that period, a missing value is set to zero. Panel F reports variables used as proxies for growth opportunities and distress. The book-to-market ratio is the log of the ratio of book equity to market equity. External finance (*EF*) is equal to the change in assets (Item 6) less the change in retained earnings (Item 36). When the change in retained earnings is not available we use net income (Item 172) less common dividends (Item 21) instead. Sales growth decile is formed using NYSE breakpoints for sales growth. Sales growth is the percentage change in net sales (Item 12). In Panels C through F, accounting data from the fiscal year ending in $t-1$ are matched to monthly returns from July of year t through June of year $t+1$. All variables are Winsorized at 99.5 and 0.5%.

	Full Sample					Subsample Means				
	N	Mean	SD	Min	Max	1960s	1970s	1980s	1990s	2000 − 1
Panel A: Returns										
R_t (%)	1,600,383	1.39	18.11	−98.13	2,400.00	1.08	1.56	1.25	1.46	1.28
MOM_{t-1} (%)	1,600,383	13.67	58.13	−85.56	343.90	21.62	12.24	15.02	13.06	11.02
Panel B: Size, Age, and Risk										
ME_{t-1} ($M)	1,600,383	621	2,319	1	23,302	388	238	395	862	1,438
Age_t (Years)	1,600,383	13.36	13.41	0.03	68.42	15.90	12.62	13.61	13.26	13.47
σ_{t-1} (%)	1,574,981	13.70	8.73	0.00	60.77	9.44	12.51	13.32	13.89	19.55
Panel C: Profitability										
$E+/BE_{t-1}$ (%)	1,600,383	10.70	10.03	0.00	65.14	12.10	12.05	11.37	9.54	9.49
$E > 0_{t-1}$	1,600,383	0.78	0.41	0.00	1.00	0.95	0.91	0.78	0.71	0.68
Panel D: Dividend Policy										
D/BE_{t-1} (%)	1,600,383	2.08	2.98	0.00	17.94	4.42	2.75	2.11	1.58	1.43
$D > 0_{t-1}$	1,600,383	0.48	0.50	0.00	1.00	0.77	0.66	0.50	0.37	0.33
Panel E: Tangibility										
PPE/A_{t-1} (%)	1,476,109	54.66	37.15	0.00	187.69	70.21	59.14	55.49	51.28	45.49
RD/A_{t-1} (%)	1,452,840	2.97	7.27	0.00	54.75		1.22	2.29	3.86	4.68
Panel F: Growth Opportunities and Distress										
BE/ME_{t-1}	1,600,383	0.94	0.86	0.02	5.90	0.70	1.37	0.95	0.76	0.82
EF/A_{t-1} (%)	1,549,817	11.44	24.24	−71.23	127.30	7.17	6.45	10.59	13.97	17.71
GS_{t-1} (Decile)	1,529,508	5.94	3.16	1.00	10.00	5.67	5.66	6.01	6.08	5.91

Table I: Summary statistics, 1963–2001.

Table II

Investor Sentiment Data, 1962–2000

Means, standard deviations, and correlations for measures of investor sentiment. In the first panel, we present raw sentiment proxies. The first (*CEFD*) is the year-end, value-weighted average discount on closed-end mutual funds. The data on prices and net asset values (NAVs) come from Neal and Wheatley (1998) for 1962 through 1993, CDA/Wiesenberger for 1994 through 1998, and turn-of-the-year issues of the *Wall Street Journal* for 1999 and 2000. The second measure (*TURN*) is detrended natural log turnover. Turnover is the ratio of reported share volume to average shares listed from the NYSE Fact Book. We detrend using the past 5-year average. The third measure (*NIPO*) is the annual number of initial public offerings. The fourth measure (*RIPO*) is the average annual first-day returns of initial public offerings. Both IPO series come from Jay Ritter, updating data analyzed in Ibbotson, Sindelar, and Ritter (1994). The fifth measure (S) is gross annual equity issuance divided by gross annual equity plus debt issuance from Baker and Wurgler (2000). The sixth measure (P^{D-ND}) is the year-end log ratio of the value-weighted average market-to-book ratios of payers and nonpayers from Baker and Wurgler (2004). Turnover, the average annual first-day return, and the dividend premium are lagged 1 year relative to the other three measures. *SENTIMENT* is the first principal component of the six sentiment proxies. In the second panel, we regress each of the six proxies on the growth in industrial production, the growth in durable, nondurable, and services consumption, the growth in employment, and a flag for NBER recessions. The orthogonalized proxies, labeled with a "$^{\perp}$," are the residuals from these regressions. $SENTIMENT^{\perp}$ is the first principal component of the six orthogonalized proxies. Superscripts a, b, and c denote statistical significance at the 1%, 5%, and 10% level, respectively.

					Correlations with Sentiment		Correlations with Sentiment Components					
	Mean	SD	Min	Max	*SENTIMENT*	$SENTIMENT^{\perp}$	*CEFD*	*TURN*	*NIPO*	*RIPO*	S	P^{D-ND}
Panel A: Raw Data												
$CEFD_t$	9.03	8.12	−10.41	23.70	-0.71^{a}	-0.60^{a}	1.00					
$TURN_{t-1}$	11.99	18.27	−26.70	42.96	0.71^{a}	0.68^{a}	-0.29^{c}	1.00				
$NIPO_t$	358.41	262.76	9.00	953.00	0.74^{a}	0.66^{a}	-0.55^{a}	0.38^{b}	1.00			
$RIPO_{t-1}$	16.94	14.93	−1.67	69.53	0.76^{a}	0.80^{a}	-0.42^{a}	0.50^{a}	0.35^{b}	1.00		
S_t	19.53	8.34	7.83	43.00	0.33^{b}	0.44^{a}	−0.01	0.30^{c}	0.16	0.26	1.00	
P^{D-ND}_{t-1}	0.20	18.67	−33.17	36.06	-0.83^{a}	-0.76^{a}	0.52^{a}	-0.50^{a}	-0.56^{a}	-0.58^{a}	−0.12	1.00
Panel B: Controlling for Macroeconomic Conditions												
$CEFD^{\perp}_t$	0.00	6.25	−18.32	9.60	-0.62^{a}	-0.63^{a}	1.00					
$TURN^{\perp}_{t-1}$	0.00	15.49	−26.03	26.37	0.69^{a}	0.71^{a}	−0.26	1.00				
$NIPO^{\perp}_t$	0.00	226.30	−435.98	484.15	0.73^{a}	0.74^{a}	-0.45^{a}	0.39^{b}	1.00			
$RIPO^{\perp}_{t-1}$	0.00	14.31	−23.55	46.54	0.77^{a}	0.83^{a}	-0.46^{a}	0.53^{a}	0.44^{a}	1.00		
$S^{\perp}_t$	0.00	6.15	−12.17	14.29	0.55^{a}	0.67^{a}	-0.41^{a}	0.32^{b}	0.50^{a}	0.47^{a}	1.00	
$P^{D-ND\perp}_{t-1}$	0.00	16.89	−43.20	35.96	-0.78^{a}	-0.77^{a}	0.26	-0.60^{a}	-0.46^{a}	-0.68^{a}	-0.28^{c}	1.00

Table II: Investor sentiment data, 1962–2000.

5.3 Tables III and IV: conditional portfolio returns and characteristic correlations

Read: Table III.

- When $\text{SENTIMENT}^{\perp}$ is negative, the bottom market-equity decile earns 2.37% per month while the top decile earns 0.92%, a strong small-stock effect.
- When $\text{SENTIMENT}^{\perp}$ is positive, the size effect largely disappears: the bottom and top deciles earn 0.73% and 0.98%.
- High-volatility stocks earn very high returns after low sentiment but very low returns after high sentiment.
- Unprofitable firms, nonpayers, and extreme growth/distress stocks show the same sentiment-conditional pattern.

Read: Table IV.

- Table IV shows that many long–short characteristic portfolios are strongly correlated.
- Small, young, volatile, unprofitable, non-dividend-paying, R&D-intensive, and extreme-growth/distress portfolios overlap substantially.
- The table motivates the regression tests with factor controls: the conditional patterns are not independent one-variable anomalies.

Economic message: Table III is the core nonparametric evidence. Table IV warns that the effects are linked across characteristics and should be interpreted as a broad speculative-stock category rather than isolated anomalies.

Additional interpretation: A major insight is that unconditional averages can wash out the evidence. If high- and low-sentiment periods generate opposite cross-sectional patterns, the unconditional return spread can look small even when sentiment is economically powerful.

Table III

Future Returns by Sentiment Index and Firm Characteristics, 1963–2001

For each month, we form 10 equal-weighted portfolios according to the NYSE breakpoints of firm size (*ME*), *age*, total risk, earnings-book ratio for profitable firms (*E/BE*), dividend-book ratio for dividend payers (*D/BE*), fixed assets (*PPE/A*), research and development (*RD/A*), book-to-market ratio (*BE/ME*), external finance over assets (*EF/A*), and sales growth (*GS*). We also calculate portfolio returns for unprofitable firms, nonpayers, zero-PP&E firms, and zero-R&D firms. We then report average portfolio returns over months in which $SENTIMENT^{\perp}$ from the previous year-end is positive, months in which it is negative, and the difference between these two averages. $SENTIMENT^{\perp}$ is positive for 1968–1970, 1972, 1979–1987, 1994, 1996–1997, and 1999–2001 (the return series ends in 2001, so the last value used is 2000).

		Decile											Comparisons			
	$SENTIMENT^{\perp}_{t-1}$	≤0	1	2	3	4	5	6	7	8	9	10	10 − 1	10 − 5	5 − 1	>0–≤0
ME	Positive		0.73	0.74	0.85	0.83	0.92	0.84	1.06	0.99	1.02	0.98	0.26	0.06	0.20	
	Negative		2.37	1.68	1.66	1.51	1.67	1.35	1.26	1.25	1.05	0.92	−1.45	−0.75	−0.70	
	Difference		−1.65	−0.93	−0.81	−0.68	−0.75	−0.51	−0.20	−0.26	−0.03	0.06	1.71	0.81	0.90	
Age	Positive		0.25	0.83	0.94	0.95	1.18	1.19	0.96	1.18	1.09	1.11	0.85	−0.07	0.93	
	Negative		1.77	1.88	1.97	1.68	1.70	1.68	1.38	1.34	1.36	1.24	−0.54	−0.46	−0.08	
	Difference		−1.52	−1.05	−1.03	−0.74	−0.51	−0.49	−0.42	−0.16	−0.27	−0.13	1.39	0.39	1.00	
σ	Positive		1.44	1.41	1.25	1.20	1.24	1.08	1.01	0.88	0.75	0.30	−1.14	−0.94	−0.20	
	Negative		1.01	1.17	1.26	1.37	1.52	1.61	1.65	1.83	2.08	2.41	1.40	0.89	0.51	
	Difference		0.43	0.24	−0.01	−0.16	−0.28	−0.53	−0.65	−0.95	−1.33	−2.11	−2.54	−1.84	−0.71	
E/BE	Positive	0.35	0.68	0.85	0.86	0.89	0.92	0.88	0.92	1.05	1.10	0.93	0.24	0.01	0.24	0.61
	Negative	2.59	2.24	2.10	2.26	1.82	1.65	1.79	1.62	1.59	1.43	1.57	−0.67	−0.08	−0.59	−0.95
	Difference	−2.25	−1.56	−1.25	−1.40	−0.93	−0.73	−0.91	−0.70	−0.54	−0.34	−0.65	0.91	0.09	0.82	1.56
D/BE	Positive	0.44	1.08	1.09	1.29	1.11	1.24	1.17	1.31	1.24	1.19	1.15	0.07	−0.09	0.16	0.75
	Negative	2.32	1.87	1.63	1.59	1.51	1.38	1.30	1.20	1.12	1.16	1.18	−0.69	−0.19	−0.49	−0.89
	Difference	−1.88	−0.79	−0.54	−0.30	−0.40	−0.14	−0.14	0.11	0.12	0.03	−0.03	0.76	0.11	0.65	1.64
PPE/A	Positive	1.31	0.48	0.66	0.74	0.81	1.04	0.90	0.79	0.87	1.04	1.05	0.57	0.02	0.56	−0.53
	Negative	1.26	1.93	1.96	1.90	1.87	1.82	1.89	1.66	1.56	1.29	1.62	−0.31	−0.20	−0.11	0.53
	Difference	0.05	−1.45	−1.31	−1.17	−1.07	−0.78	−0.99	−0.87	−0.69	−0.25	−0.56	0.88	0.22	0.67	−1.05
RD/A	Positive	0.80	1.21	1.04	1.37	1.37	1.34	1.22	1.24	1.29	1.39	1.38	0.17	0.04	0.13	0.55
	Negative	1.63	1.57	1.47	1.58	1.73	1.66	1.81	1.97	2.04	2.13	2.44	0.87	0.78	0.09	0.43
	Difference	−0.83	−0.36	−0.43	−0.22	−0.36	−0.32	−0.60	−0.73	−0.75	−0.74	−1.05	−0.69	−0.74	0.04	0.12
BE/ME	Positive		0.03	0.61	0.82	0.87	0.96	1.09	1.17	1.18	1.29	1.27	1.24	0.31	0.93	
	Negative		1.41	1.43	1.46	1.54	1.61	1.69	1.87	1.94	2.18	2.45	1.04	0.84	0.20	
	Difference		−1.38	−0.81	−0.64	−0.67	−0.65	−0.60	−0.70	−0.76	−0.88	−1.18	0.20	−0.53	0.73	
EF/A	Positive		1.08	1.04	1.25	1.18	1.19	1.17	1.02	0.92	0.75	−0.01	−1.09	−1.20	0.11	
	Negative		2.43	2.09	1.85	1.75	1.59	1.53	1.51	1.51	1.71	1.53	−0.90	−0.06	−0.84	
	Difference		−1.35	−1.05	−0.59	−0.57	−0.40	−0.35	−0.49	−0.60	−0.96	−1.54	−0.18	−1.14	0.96	
GS	Positive		0.70	1.07	1.19	1.15	1.21	1.18	1.22	1.10	0.81	0.05	−0.65	−1.16	0.51	
	Negative		2.49	1.78	1.61	1.54	1.47	1.57	1.68	1.78	1.68	1.69	−0.80	0.22	−1.02	
	Difference		−1.79	−0.71	−0.42	−0.40	−0.26	−0.39	−0.46	−0.68	−0.87	−1.64	0.15	−1.38	1.53	

Table III: Future returns by sentiment index and firm characteristics.

Table IV

Correlations of Portfolio Returns, 1963–2001

Correlations among characteristics-based portfolios. The sample period includes monthly returns from 1963 to 2001. The long–short portfolios are formed based on firm characteristics: firm size (*ME*), *age*, total risk (σ), profitability (*E*), dividends (*D*), fixed assets (*PPE*), research and development (*RD*), book-to-market ratio (*BE/ME*), external finance over assets (*EF/A*), and sales growth decile (*GS*). High is defined as a firm in the top three NYSE deciles, low is defined as a firm in the bottom three NYSE deciles, and medium is defined as a firm in the middle four NYSE deciles. Superscripts a, b, and c denote statistical significance at the 1%, 5%, and 10%, level, respectively.

		Size, Age, and Risk			Profitability, Dividends		Tangibility		Growth Opportunities and Distress			Growth Opportunities			Distress		
		ME	*Age*	σ	*E*	*D*	*PPE/A*	*RD/A*	*BE/ME*	*EF/A*	*GS*	*BE/ME*	*EF/A*	*GS*	*BE/ME*	*EF/A*	*GS*
ME	*SMB*	1.00															
Age	High-Low	−0.78^{a}	1.00														
σ	High-Low	0.77^{a}	−0.92^{a}	1.00													
E	>0 − <0	−0.64^{a}	0.85^{a}	−0.85^{a}	1.00												
D	>0 − =0	−0.76^{a}	0.94^{a}	−0.96^{a}	0.91^{a}	1.00											
PPE/A	High-Low	−0.74^{a}	0.85^{a}	−0.87^{a}	0.71^{a}	0.84^{a}	1.00										
RD/A	High-Low	0.61^{a}	−0.78^{a}	0.82^{a}	−0.69^{a}	−0.80^{a}	−0.79^{a}	1.00									
BE/ME	*HML*	0.09^{a}	0.47^{a}	−0.46^{a}	0.23^{a}	0.43^{a}	0.51^{a}	−0.73^{a}	1.00								
EF/A	High-Low	0.33^{a}	−0.55^{a}	0.56^{a}	−0.34^{a}	−0.51^{a}	−0.59^{a}	0.66^{a}	−0.65^{a}	1.00							
GS	High-Low	0.14^{a}	−0.17^{a}	0.23^{a}	0.10^{b}	−0.15^{a}	−0.34^{a}	0.29^{a}	−0.57^{a}	0.73^{a}	1.00						
BE/ME	Medium-Low	−0.37^{a}	0.66^{a}	−0.68^{a}	0.48^{a}	0.64^{a}	0.61^{a}	−0.80^{a}	0.81^{a}	−0.74^{a}	−0.49^{a}	1.00					
EF/A	High-Medium	0.65^{a}	−0.87^{a}	0.90^{a}	−0.76^{a}	−0.88^{a}	−0.83^{a}	0.76^{a}	−0.51^{a}	0.77^{a}	0.42^{a}	−0.73^{a}	1.00				
GS	High-Medium	0.65^{a}	−0.84^{a}	0.88^{a}	−0.69^{a}	−0.86^{a}	−0.81^{a}	0.80^{a}	−0.62^{a}	0.78^{a}	0.55^{a}	−0.77^{a}	0.92^{a}	1.00			
BE/ME	High-Medium	0.31^{a}	−0.30^{a}	0.29^{a}	−0.50^{a}	−0.35^{a}	−0.22^{a}	−0.01	0.40^{a}	−0.31^{a}	−0.53^{a}	0.16^{a}	0.14^{a}	0.03	1.00		
EF/A	Medium-Low	−0.61^{a}	0.69^{a}	−0.70^{a}	0.77^{a}	0.74^{a}	0.56^{a}	−0.39^{a}	−0.01	0.04	0.26^{a}	0.23^{a}	−0.60^{a}	−0.47^{a}	−0.61^{a}	1.00	
GS	Medium-Low	−0.61^{a}	0.80^{a}	−0.79^{a}	0.88^{a}	0.83^{a}	0.60^{a}	−0.62^{a}	0.17^{a}	−0.20^{a}	0.31^{a}	0.42^{a}	−0.65^{a}	−0.63^{a}	−0.53^{a}	0.77^{a}	1.00

Table IV: Correlations of characteristic-based portfolio returns.

5.4 Table V: formal return-forecasting regressions

Read:

- Table V estimates long–short portfolio return regressions on lagged sentiment with and without standard factor controls.
- The coefficients generally have the predicted signs: sentiment forecasts low future returns for speculative/hard-to-arbitrage legs relative to safer legs.
- Adding RMRF, SMB, HML, and UMD often reduces magnitudes, especially where size is important, but many results remain economically and statistically meaningful.
- Growth-opportunity and distress portfolios show that the effect is not just a standard high-minus-low value spread; both extremes of the distribution can matter.

Economic message: The regression evidence turns the visual and sort-based pattern into a formal forecasting test. It shows that the result is not simply an artifact of one sort or one high-sentiment episode.

Additional interpretation: Table V is the bridge between behavioral narrative and asset-pricing discipline. It asks whether sentiment has incremental predictive power after familiar factor controls. The partial attenuation after SMB is expected: small firms are exactly where sentiment should matter. The key is that SMB does not eliminate the broader conditional pattern across age, volatility, profitability, dividends, growth, and distress.

Table V

Time Series Regressions of Portfolio Returns, 1963 to 2001

Regressions of long–short portfolio returns on lagged *SENTIMENT*, the market risk premium (*RMRF*), the Fama–French factors (*HML* and *SMB*), and a momentum factor (*UMD*).

$$R_{X_{it}=High,t} - R_{X_{it}=Low,t} = c + dSENTIMENT_{t-1} + \beta RMRF_t + sSMB_t + hHML_t + mUMD_t + u_t.$$

The sample period includes monthly returns from 1963 to 2001. The long–short portfolios are formed based on firm characteristics (X): firm size (*ME*), *age*, total risk (σ), profitability (E), dividends (D), fixed assets (*PPE*), research and development (*RD*), book-to-market ratio (*BE/ME*), external finance over assets (*EF/A*), and sales growth decile (*GS*). High is defined as a firm in the top three NYSE deciles, low is defined as a firm in the bottom three NYSE deciles, and medium is defined as a firm in the middle four NYSE deciles. Average monthly returns are matched to *SENTIMENT* from the previous year-end. $SENTIMENT^{\perp}$ index is based on six sentiment proxies that have been orthogonalized to growth in industrial production, the growth in durable, nondurable, and services consumption, the growth in employment, and a flag for NBER recessions; the components of *SENTIMENT* are not orthogonalized. The first and third sets of columns show univariate regression results, while the second and the fourth columns include *RMRF*, *SMB*, *HML*, and *UMD* as control variables. *SMB* (*HML*) is not included as a control variable when *SMB* (*HML*) is the dependent variable. Bootstrapped p-values are in brackets.

		$SENTIMENT_{t-1}$		$SENTIMENT_{t-1}$ Controlling for *RMRF*, *SMB*, *HML*, *UMD*		$SENTIMENT^{\perp}_{t-1}$		$SENTIMENT^{\perp}_{t-1}$ Controlling for *RMRF*, *SMB*, *HML*, *UMD*	
		d	**p(*d*)**	***d***	**p(*d*)**	***d***	**p(*d*)**	***d***	**p(*d*)**
Panel A: Size, Age, and Risk									
ME	***SMB***	−0.4	[0.04]	−0.3	[0.08]	−0.4	[0.06]	−0.3	[0.15]
Age	**High-Low**	0.5	[0.02]	0.2	[0.08]	0.5	[0.01]	0.2	[0.04]
σ	**High-Low**	−1.0	[0.01]	−0.5	[0.01]	−0.9	[0.00]	−0.4	[0.01]
Panel B: Profitability and Dividend Policy									
E	**>0 – <0**	0.8	[0.00]	0.5	[0.02]	0.8	[0.00]	0.4	[0.02]
D	**>0 – =0**	0.8	[0.00]	0.4	[0.00]	0.8	[0.00]	0.4	[0.00]

(continued)

Table V: Time-series regressions of portfolio returns, part 1.

Table V—*Continued*

		$SENTIMENT_{t-1}$		$SENTIMENT_{t-1}$ Controlling for *RMRF*, *SMB*, *HML*, *UMD*		$SENTIMENT^{\perp}_{t-1}$		$SENTIMENT^{\perp}_{t-1}$ Controlling for *RMRF*, *SMB*, *HML*, *UMD*	
		d	**p(*d*)**	***d***	**p(*d*)**	***d***	**p(*d*)**	***d***	**p(*d*)**
Panel C: Tangibility									
PPE/A	**High-Low**	0.4	[0.12]	0.1	[0.65]	0.4	[0.07]	0.1	[0.53]
RD/A	**High-Low**	−0.3	[0.25]	0.0	[0.77]	−0.3	[0.24]	0.1	[0.62]
Panel D: Growth Opportunities and Distress									
BE/ME	***HML***	0.1	[0.47]	0.0	[1.00]	0.2	[0.25]	0.1	[0.68]
EF/A	**High-Low**	−0.1	[0.24]	−0.1	[0.46]	−0.2	[0.13]	−0.1	[0.40]
GS	**High-Low**	−0.1	[0.53]	−0.0	[0.67]	−0.1	[0.23]	−0.1	[0.51]
Panel E: Growth Opportunities									
BE/ME	**Medium-Low**	0.2	[0.11]	0.1	[0.39]	0.3	[0.06]	0.1	[0.35]
EF/A	**High-Medium**	−0.4	[0.00]	−0.2	[0.01]	−0.4	[0.00]	−0.2	[0.00]
GS	**High-Medium**	−0.4	[0.00]	−0.2	[0.00]	−0.4	[0.00]	−0.2	[0.00]
Panel F: Distress									
BE/ME	**High-Medium**	−0.1	[0.21]	−0.1	[0.49]	−0.1	[0.30]	−0.1	[0.53]
EF/A	**Medium-Low**	0.3	[0.00]	0.2	[0.01]	0.2	[0.00]	0.2	[0.00]
GS	**Medium-Low**	0.3	[0.01]	0.2	[0.04]	0.3	[0.01]	0.2	[0.07]

Table V: Time-series regressions of portfolio returns, part 2.

5.5 Tables VI and VII: earlier data and conditional beta tests

Read: Table VI.

- The paper extends some tests back to 1935 using the subset of sentiment proxies and characteristics available historically.
- The volatility and dividend patterns remain especially strong over the longer period.
- The earlier data are less rich, so Table VI is a robustness check rather than the main identification device.

Read: Table VII.

- Table VII tests whether the sentiment-conditioned return patterns can be explained by sentiment-conditioned market betas.
- Most beta interaction coefficients do not line up with the return predictability in Table V.
- Some significant beta interactions appear, but several have the wrong sign relative to the risk explanation.

Economic message: The paper is careful about the risk alternative. Table VI asks whether the patterns survive in earlier data. Table VII asks whether changing market betas explain them. The evidence favors mispricing or expectational error over a simple conditional-beta story.

Additional interpretation: The beta test is conceptually important because sentiment-based predictability could be reinterpreted as risk compensation. But a risk explanation would require the sign of risk premia or relative risk exposure to flip across sentiment states in unintuitive ways. Table VII makes that explanation less persuasive.

Table VI

Time Series Regressions of Portfolio Returns, 1935 to 2001

Regressions of long–short portfolio returns on lagged *SENTIMENT*, the market risk premium (*RMRF*), the Fama–French factors (*HML* and *SMB*), and a momentum factor (*UMD*).

$$R_{X_{it}=High,t} - R_{X_{it}=Low,t} = c + dSENTIMENT_{t-1} + \beta RMRF_t + sSMB_t + hHML_t + mUMD_t + u_t.$$

The long–short portfolios are formed based on firm characteristics (X): firm size (*ME*), *age*, total risk (σ), and dividends (D). High is defined as a firm in the top three NYSE deciles, and low is defined as a firm in the bottom three NYSE deciles. The sentiment index is the first principal component of the closed-end fund discount (*CEFD*), the equity share (S), and the lag of detrended log turnover (*TURN*). Average monthly returns are matched to *SENTIMENT* from the previous year-end. $SENTIMENT^{\perp}$ index is based on six sentiment proxies that have been orthogonalized to growth in industrial production, the growth in durable, nondurable, and services consumption, the growth in employment, and a flag for NBER recessions; the components of *SENTIMENT* are not orthogonalized. The first and third sets of columns show univariate regression results, while the second and the fourth columns include *RMRF*, *SMB*, *HML*, and *UMD* as control variables. *SMB* (*HML*) is not included as a control variable when *SMB* (*HML*) is the dependent variable. Bootstrapped p-values are in brackets.

		$SENTIMENT_{t-1}$		$SENTIMENT_{t-1}$ Controlling for *RMRF*, *SMB*, *HML*, *UMD*		$SENTIMENT^{\perp}_{t-1}$		$SENTIMENT^{\perp}_{t-1}$ Controlling for *RMRF*, *SMB*, *HML*, *UMD*	
		d	**p(*d*)**	***d***	**p(*d*)**	***d***	**p(*d*)**	***d***	**p(*d*)**
				Panel A: 1935–2001					
ME	***SMB***	−0.3	[0.03]	−0.2	[0.07]	−0.3	[0.04]	−0.2	[0.07]
Age	**High-Low**	0.2	[0.18]	0.1	[0.38]	0.2	[0.10]	0.1	[0.26]
σ	**High-Low**	−1.0	[0.00]	−0.4	[0.00]	−0.8	[0.00]	−0.4	[0.00]
D	$> \mathbf{0} - = \mathbf{0}$	0.9	[0.00]	0.5	[0.01]	0.7	[0.00]	0.4	[0.01]
				Panel B: 1935–1961					
ME	**SMB**	−0.3	[0.05]	−0.3	[0.05]	−0.1	[0.33]	−0.1	[0.33]
Age	**High-Low**	−0.1	[0.41]	−0.1	[0.41]	−0.1	[0.04]	−0.1	[0.05]
σ	**High-Low**	−0.8	[0.01]	−0.8	[0.01]	−0.4	[0.14]	−0.4	[0.16]
D	$>\mathbf{0} - = \mathbf{0}$	0.9	[0.01]	0.9	[0.01]	0.5	[0.10]	0.5	[0.11]

recent data.[14] One possibility for the insignificant results on the *Age* portfolio is that we measure age as the number of months for which CRSP data are available. Anecdotal evidence suggests that in these early data, there are fewer truly "young" firms listing on the NYSE. In contrast, in recent years, many genuinely young IPOs start trading on Nasdaq, so our way of measuring age may be more meaningful.

The longer time series make it possible to conduct an out-of-sample test. In unreported results, we compare the in-sample reduction in root mean squared

[14] For a more detailed look at earlier data, see Gruber (1966). He documents changes in the cross-sectional determinants of stock prices between 1951 and 1963, and argues that they are connected to changes in the average investor's time horizon, that is, shifts in the term structure of discount rates. This is similar to our notion of sentiment as a shift in the propensity to speculate.

(a) Table VI: Time-series regressions, 1935–2001.

Table VII

Conditional Market Betas, 1963–2001

Regressions of long–short portfolio returns on the market risk premium (*RMRF*) and the market risk premium interacted with *SENTIMENT*.

$$R_{X_{it}=High,t} - R_{X_{it}=Low,t} = c + dSENTIMENT_{t-1} + \beta(e + fSENTIMENT_{t-1})RMRF_t + u_t.$$

The long–short portfolios are formed based on firm characteristics (X): firm size (*ME*), *age*, total risk (σ), profitability (E), dividends (D), fixed assets (*PPE*), research and development (*RD*), book-to-market ratio (*BE/ME*), external finance over assets (*EF/A*), and sales growth decile (*GS*). High is defined as a firm in the top three NYSE deciles, low is defined as a firm in the bottom three NYSE deciles, and medium is defined as a firm in the middle four NYSE deciles. Monthly returns are matched to *SENTIMENT* from the previous year-end. $SENTIMENT^{\perp}$ index is based on six sentiment proxies that have been orthogonalized to growth in industrial production, the growth in durable, nondurable and services consumption, the growth in employment and a flag for NBER recessions; the components of *SENTIMENT* are not orthogonalized. Heteroskedasticity-robust p-values are in brackets. A superscript "a" indicates a statistically significant βf that matches the sign of the return predictability from Table V; "b" indicates a statistically significant βf that does not match.

		$SENTIMENT_{t-1}$		$SENTIMENT^{\perp}_{t-1}$	
		βf	$t(\beta f)$	βf	$t(\beta f)$
		Panel A: Size, Age, and Risk			
ME	***SMB***	−0.03	[0.48]	−0.02	[0.62]
Age	**High-Low**	−0.05	[0.19]	−0.07	[0.09]
σ	**High-Low**	0.00	[0.98]	0.03	[0.58]
		Panel B: Profitability and Dividend Policy			
E	$>\mathbf{0} - <\mathbf{0}$	−0.04	[0.47]	−0.00	[0.98]
D	$>\mathbf{0} - = \mathbf{0}$	−0.01	[0.75]	−0.04	[0.35]
		Panel C: Tangibility			
PPE/A	**High-Low**	−0.00	[0.94]	−0.01	[0.76]
RD/A	**High-Low**	0.12[b]	[0.01]	0.17[b]	[0.00]
		Panel D: Growth Opportunities and Distress			
BE/ME	***HML***	−0.10[b]	[0.02]	−0.12[b]	[0.00]
EF/A	**High-Low**	0.06[b]	[0.00]	0.07[b]	[0.00]
GS	**High-Low**	0.42	[0.06]	0.37	[0.08]
		Panel E: Growth Opportunities			
BE/ME	**Medium-Low**	−0.06	[0.05]	−0.09[b]	[0.01]
EF/A	**High-Medium**	0.03	[0.13]	0.04	[0.05]
GS	**High-Medium**	0.05[b]	[0.02]	0.06[b]	[0.01]
		Panel F: Distress			
BE/ME	**High-Medium**	−0.07[a]	[0.00]	−0.07[a]	[0.00]
EF/A	**Medium-Low**	0.03	[0.08]	0.02	[0.17]
GS	**Medium-Low**	−0.01	[0.62]	−0.02	[0.27]

(b) Table VII: Conditional market betas, 1963–2001.

5.6 Table VIII: earnings-announcement evidence

Read:

- Table VIII regresses quarterly earnings-announcement effects on lagged orthogonalized sentiment across characteristic deciles.
- Some deciles show announcement-return effects that line up with the return-predictability results.
- The evidence suggests that at least part of the sentiment effect works through mistaken expectations about fundamentals.

Economic message: If sentiment-driven overpricing reflects overly optimistic beliefs, bad news should be revealed when firms announce earnings. Table VIII provides supportive but necessarily partial evidence for this expectational-error channel.

Additional interpretation: The test is a lower bound. Some mispricing may correct through other news, gradual price pressure, issuance events, analyst revisions, or changes in market-wide appetite for speculative stocks. Therefore, imperfect announcement evidence does not weaken the main result; it simply shows that earnings news is one observable correction mechanism.

Table VIII

Earnings Announcement Effects, 1973–2001

Regressions of average quarterly earnings announcement effects on lagged *SENTIMENT*.

$$CAR_{X_{it}=Decile,t} = c + dSENTIMENT^{\perp}_{t-1} + u_t.$$

We report the coefficient b below. For each calendar quarter, we form 10 portfolios according to the NYSE breakpoints of firm size (*ME*), *age*, total risk, earnings-book ratio for profitable firms (*E/BE*), dividend-book ratio for dividend payers (*D/BE*), fixed assets (*PPE/A*), research and development (*RD/A*), book-to-market ratio (*BE/ME*), external finance over assets (*EF/A*), and sales growth (*GS*). We also calculate average announcement effects for unprofitable firms, nonpayers, zero-PP&E firms, and zero-R&D firms. Quarterly average announcement effects are matched to $SENTIMENT^{\perp}$ from the previous year-end. $SENTIMENT^{\perp}$ index is based on six sentiment proxies that have been orthogonalized to growth in industrial production, the growth in durable, nondurable, and services consumption, the growth in employment, and a flag for NBER recessions. Heteroskedasticity-robust p-values are in brackets.

	Decile										
	≤0	1	2	3	4	5	6	7	8	9	10
ME		−0.18	−0.05	−0.02	−0.07	−0.02	−0.05	0.00	−0.03	0.01	−0.03
		[0.03]	[0.35]	[0.76]	[0.27]	[0.64]	[0.36]	[0.99]	[0.50]	[0.90]	[0.48]
Age		−0.11	0.02	−0.06	−0.06	−0.12	0.00	0.06	−0.03	0.00	−0.12
		[0.20]	[0.79]	[0.27]	[0.35]	[0.08]	[1.00]	[0.21]	[0.64]	[1.00]	[0.01]
σ		0.10	0.05	0.05	0.05	−0.04	−0.03	0.03	−0.02	−0.08	−0.30
		[0.05]	[0.23]	[0.26]	[0.32]	[0.45]	[0.59]	[0.69]	[0.79]	[0.27]	[0.00]
E/BE	−0.31	−0.24	0.01	0.11	−0.08	−0.09	−0.02	−0.01	0.04	0.04	0.10
	[0.00]	[0.01]	[0.92]	[0.24]	[0.31]	[0.23]	[0.73]	[0.85]	[0.54]	[0.47]	[0.08]
D/BE	−0.18	0.00	−0.05	−0.02	0.01	0.00	0.03	−0.03	−0.08	0.04	−0.10
	[0.02]	[0.94]	[0.43]	[0.72]	[0.80]	[1.00]	[0.56]	[0.60]	[0.07]	[0.43]	[0.16]
PPE/A	−0.12	−0.12	−0.12	−0.14	−0.02	0.01	0.00	−0.12	−0.10	−0.06	−0.05
	[0.14]	[0.12]	[0.14]	[0.05]	[0.75]	[0.87]	[0.97]	[0.07]	[0.09]	[0.28]	[0.41]
RD/A	−0.05	−0.10	−0.07	−0.04	−0.10	−0.15	−0.28	−0.29	−0.14	−0.11	−0.08
	[0.32]	[0.45]	[0.46]	[0.66]	[0.22]	[0.12]	[0.00]	[0.01]	[0.10]	[0.21]	[0.42]
BE/ME		−0.11	−0.03	0.03	−0.04	−0.04	0.04	−0.05	0.01	−0.14	−0.12
		[0.06]	[0.67]	[0.62]	[0.38]	[0.43]	[0.54]	[0.37]	[0.87]	[0.05]	[0.24]
EF/A		−0.07	−0.01	0.05	−0.06	0.03	−0.04	−0.10	0.01	−0.11	−0.09
		[0.38]	[0.95]	[0.40]	[0.31]	[0.57]	[0.51]	[0.07]	[0.90]	[0.06]	[0.20]
GS		−0.20	0.00	−0.08	−0.02	−0.03	0.04	−0.03	0.03	0.00	−0.11
		[0.02]	[0.99]	[0.23]	[0.68]	[0.60]	[0.51]	[0.56]	[0.60]	[0.94]	[0.15]

Table VIII: Earnings announcement effects, 1973–2001.

6 Short summary

- The paper argues that investor sentiment affects stocks differentially, not uniformly.
- Sentiment-sensitive stocks are hard to value and hard to arbitrage: small, young, volatile, unprofitable, nonpayers, distressed firms, and extreme-growth firms.
- The authors build a sentiment index from closed-end fund discounts, turnover, IPO volume, IPO first-day returns, equity issuance share, and dividend premium.
- The orthogonalized sentiment index removes macro conditions and still behaves similarly to the raw index.
- Portfolio sorts show that speculative stocks have high subsequent returns after low sentiment and low subsequent returns after high sentiment.
- Regression tests confirm the patterns after controlling for Fama-French and momentum factors.
- Conditional beta tests do not provide a satisfying risk-based explanation.
- Earnings-announcement tests suggest that sentiment effects partly reflect errors in expectations about fundamentals.
- The result is a conditional characteristics model: expected returns depend on both firm characteristics and market sentiment.

7 Expanded conclusion

7.1 Core conclusion

Baker and Wurgler show that investor sentiment has cross-sectional pricing effects. It does not merely move the whole market up or down. It changes the relative valuation of speculative and bond-like firms. When sentiment is high, stocks that investors find exciting and arbitrageurs find difficult become relatively overpriced and subsequently underperform. When sentiment is low, the same stocks are relatively cheap and subsequently outperform.

7.2 Asset-pricing implication

The paper challenges a purely classical view in which expected returns are fully summarized by systematic risk. It does not deny risk; rather, it shows that a risk-only interpretation must explain why the sign and strength of characteristic premiums change with beginning-of-period sentiment. This is a high bar. A conditional characteristics model with sentiment explains the patterns more directly.

7.3 Behavioral-finance implication

The paper gives empirical content to the phrase "sentiment." Sentiment is not simply optimism in survey data. It appears in trading volume, IPO markets, equity issuance, closed-end fund discounts, and the relative valuation of dividend payers. The common component across these variables captures waves of speculative demand.

7.4 Portfolio implication

The findings imply that style tilts are time-varying. A portfolio tilted toward small, young, volatile, unprofitable, or non-dividend-paying stocks has very different expected-return implications when sentiment is high than when sentiment is low. For investors, sentiment is therefore a state variable for style timing, although implementation requires careful attention to transaction costs, shorting constraints, and data-mining risk.

7.5 Corporate-finance implication

The paper also links asset pricing to corporate finance. If sentiment overvalues certain firm characteristics, firms may respond by issuing equity, going public, investing in growth narratives, or catering to investor demand. This connects the cross-section of returns to market timing, IPO waves, dividend catering, and the supply of securities.

7.6 Interpretation of the risk debate

The risk explanation is not impossible, but it must be complicated. It needs time-varying risk prices or betas that reverse relative rankings across sentiment states. The conditional beta and earnings-announcement evidence makes the behavioral interpretation more natural. The paper therefore shifts the burden of proof: sentiment-based mispricing is not a residual story after risk is exhausted; it is a structured, testable explanation for predictable cross-sectional patterns.

7.7 What future work should do

- Test the sentiment-conditioned characteristic patterns in international markets and post-2001 samples.
- Measure sentiment with modern data such as retail flows, social media, options demand, and brokerage-level trading records.
- Separate belief errors from limits to arbitrage by studying short-sale constraints, lending fees, and institutional ownership.
- Connect sentiment to corporate actions such as issuance, repurchases, IPO timing, investment, and dividend policy.
- Build asset-pricing models where sentiment is a priced state variable or a non-priced mispricing state variable that interacts with characteristics.

7.8 Final takeaway

The lasting contribution is the interaction between sentiment and characteristics. The paper does not simply say that investor sentiment exists. It shows where sentiment should matter, how to measure it, and how to test it. The most important lesson is that the cross-section is conditional: the same firm characteristic can mean very different expected returns depending on the market's speculative mood.